

Leakage

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sampling and Intended Use

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Inspect Before Modeling

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features, Labels, Targets

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Distinguish frames from groups for Splits and related examples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Each of five plants has four photos. A random photo split puts different views of every plant in both training and test sets. Explain the problem and repair.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected grouping rule.

6. Solve the given case

Ten photos per leaf are randomly split so one leaf appears in train and test.

Explain and repair leakage.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A feature records the expert's final species decision, made after the prediction time. Why is this leakage, and what should happen to it?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 8 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Audit filenames, metadata, post-diagnosis fields, and duplicate plant identities; record every removed leakage path.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected grouping rule.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the three roles

Explain what training, validation and test sets are used for. What changes if you repeatedly choose a model using its test score?

My reasoning and evidence

13. W2 Distinguish frames from groups

For the worked A through F plan, give training, validation and test counts first in groups and then in rows. Interpret [] from the code and state one problem it cannot detect.

My reasoning and evidence

14. W3 Repair a related example leak

A1 and A2 are in training and A3 is in test. Explain why unique frame IDs and a fixed random seed do not fix this. Propose a repair and what to record.

My reasoning and evidence

15. W4 Audit and repair independently

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is training P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared groups for each pair of sets. Repair it by assigning P and Q to training, R to validation and S to test. List all frames in each role and give row and group counts. State one limit of the repaired plan.

My reasoning and evidence

16. W5 Match separation to the claim

A model will be evaluated on new routes. Route North has trips A and B; route South has C and D. A proposed split trains on A and C and tests on B and D. Explain the problem and give a route-based alternative. Is this enough for a strong performance claim?

My reasoning and evidence

17. Scaling without leakage — W1 Learn the scale from training

Using A,B,C, give the minimum, maximum and range for each feature. Transform all three training rows and query (11,120). Explain which rows supplied the scale.

My reasoning and evidence

18. Scaling without leakage — W2 Trace the changed prediction

For query (11,120), calculate raw and scaled distances to A,B,C. Give both k=1 predictions. Does this prove that scaling always improves predictions?

My reasoning and evidence

19. Scaling without leakage — W3 Keep one coordinate system

Transform (12,80) using the stored training scale. Explain the negative value and why validation must not fit its own scale. Give the worked validation and baseline scores.

My reasoning and evidence

20. Scaling without leakage — W4 Scale a fresh dataset independently

New training rows are P (20,1000) clear, Q (40,600) blocked, R (40,1000) blocked. Fit minima and ranges. For query (22,680), show transformed features, raw and scaled distances to P,Q,R, and both k=1 predictions. Transform new row (50,520) with the same scale.

My reasoning and evidence

21. Scaling without leakage — W5 Repair two preprocessing mistakes

A learner fits scales using training plus validation features but no labels. Another divides by a zero training range. Explain both problems and give a recorded repair for each.

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Leakage

Split by plant identity when testing generalisation to new plants.

Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/8

Split by plant, not by view

	view1	view2	view3
Cedar	train	train	train
Elm	train	train	train
Fir	test	test	test

Read a cell using both its row and column labels.

Leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group related views

All three Cedar views stay together; likewise Elm and Fir.



2. Assign groups

Training receives six views from Cedar/Elm; test receives three Fir views.



3. State claim

The final test probes one unseen plant, not three independent plants.

CHANGE ONE THING

Move Fir's first view into training while retaining its other two in test.

Explain the changed result using the matching lesson.

How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

24. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Split by plant identity when testing generalisation to new plants.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final test probes one unseen plant, not three independent plants.

1. Assign groups
2. State claim
3. Group related views

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: State claim The final test probes one unseen plant, not three independent plants. Read the picture in order. The final test probes one unseen plant, not three independent plants. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Move Fir's first view into training while retaining its other two in test. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The unseen-plant evaluation is compromised.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The unseen-plant evaluation is compromised. Fir's identity now appears in training; different view IDs do not remove the shared plant relationship.

Repair a missing building block

Review Sampling and Intended Use (ai-4/7). Goal: Inspect row meaning missing values and repeated records. Short practice: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Features, Labels, Targets (ai-4/2). Goal: Separate prediction inputs from observed targets. Short practice: Why can a historical table contain a field that is an invalid input? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
